

Matteo Palmonari

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Beyond Query Answering: *Exploration, Profiling & Detection of Salient Information*



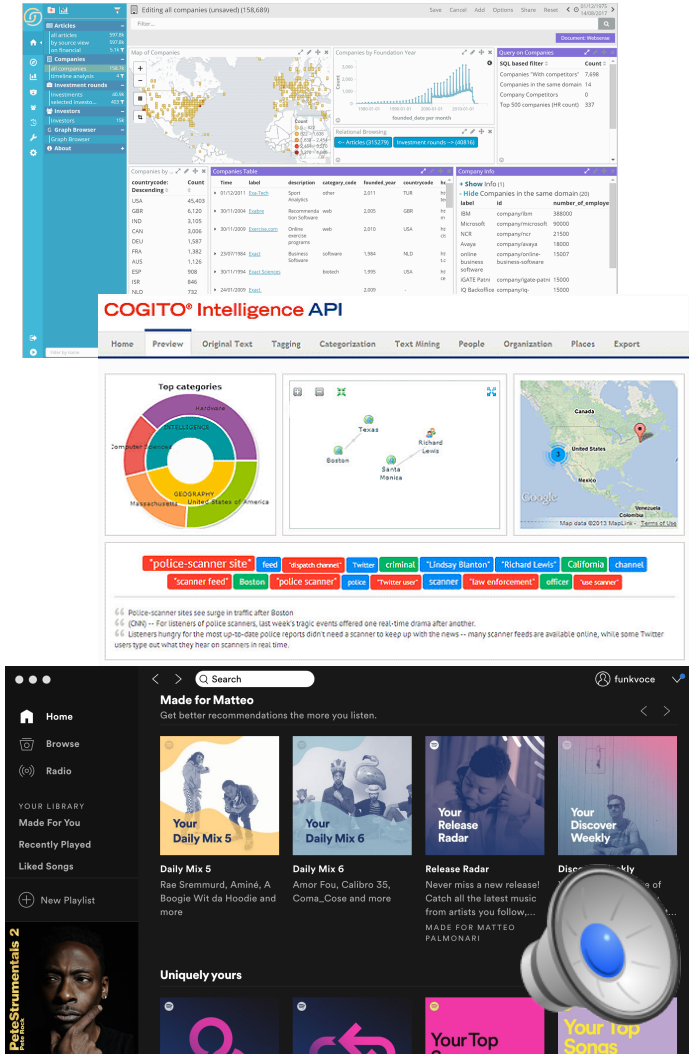
INSID&S Lab
Interaction and **S**emantics
for **I**nnovation with **D**ata & **S**ervices



Beyond Query Answering

- Structured queries represent specific information needs
 - You need to know what you are searching for to formulate the query
- Access to information beyond querying
 - **Passive** applications: support exploratory search, i.e., let users browse through the data before formulating a precise query
 - Discussed through a short presentation on supporting **exploration of large Knowledge Bases**
 - **Active** applications: push content to the user, i.e., predict what the user is interested in
 - Discussed by introducing a paradigmatic case: **recommended systems**

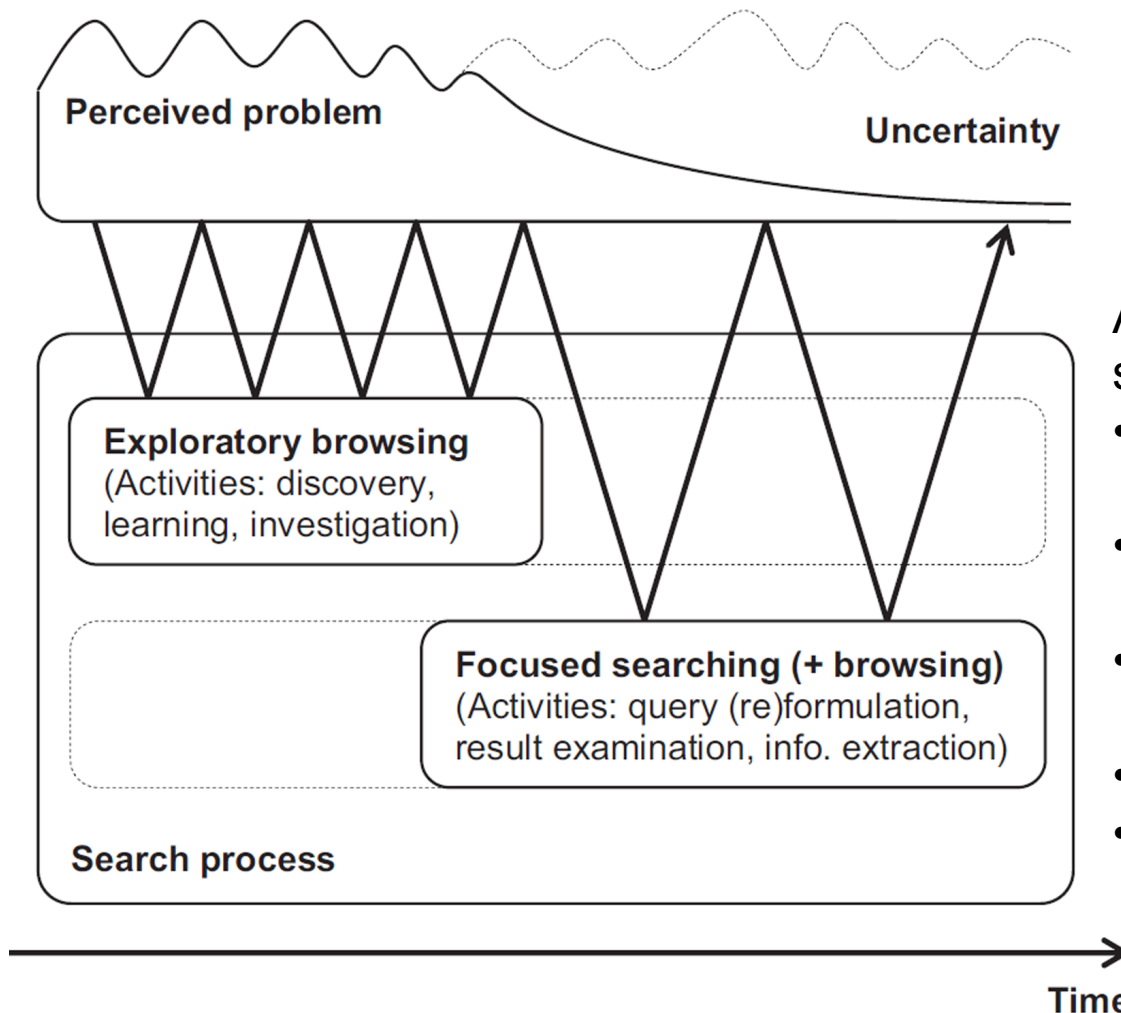
3rd important aspect of semantics:
(inference + similarity +) relevance



Exploration of Large Knowledge Bases



Beyond Query: Exploratory Search (White & Roth)



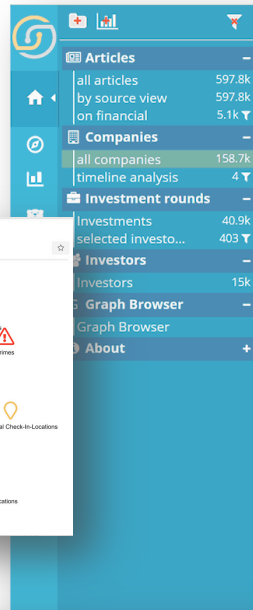
A system for exploratory search should:

- Support query refining and filtering
- Disambiguate and rank results using context
- Use visualizations to facilitate insights
- Support collaboration
- Consider user history



Ryen W White and Resa A Roth. Exploratory search: beyond the query-response paradigm (synthesis lectures on information concepts, retrieval & services). Morgan and Claypool Publishers, 3:9–23, 2009.

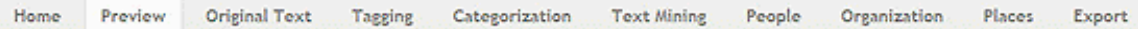
Graph



Check more info online at:
<https://siren.io/platform-overview/>



“



...one real-time drama after another.



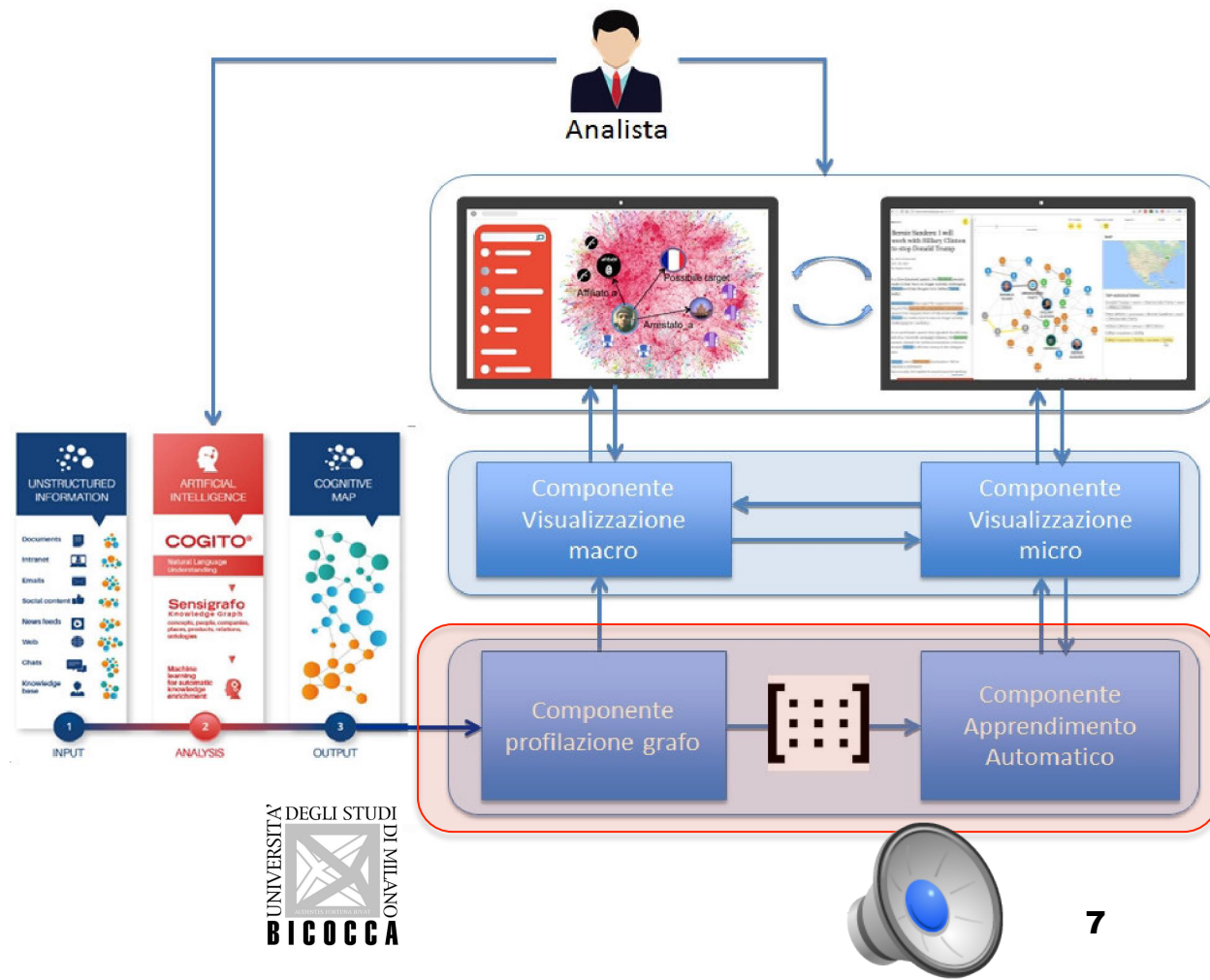
Profiling & Learning to Explore Knowledge – SMARTGraphX Project Proposal

Profiling and exploration of large KGs

- Ontology-base profiling: understand the content of the KG
- Entity-specific profiling and heuristics to define relevant elements

Learning what is more relevant for the user

Visualization: important but not sufficient ingredient



Explorative Search of Relations in KGs

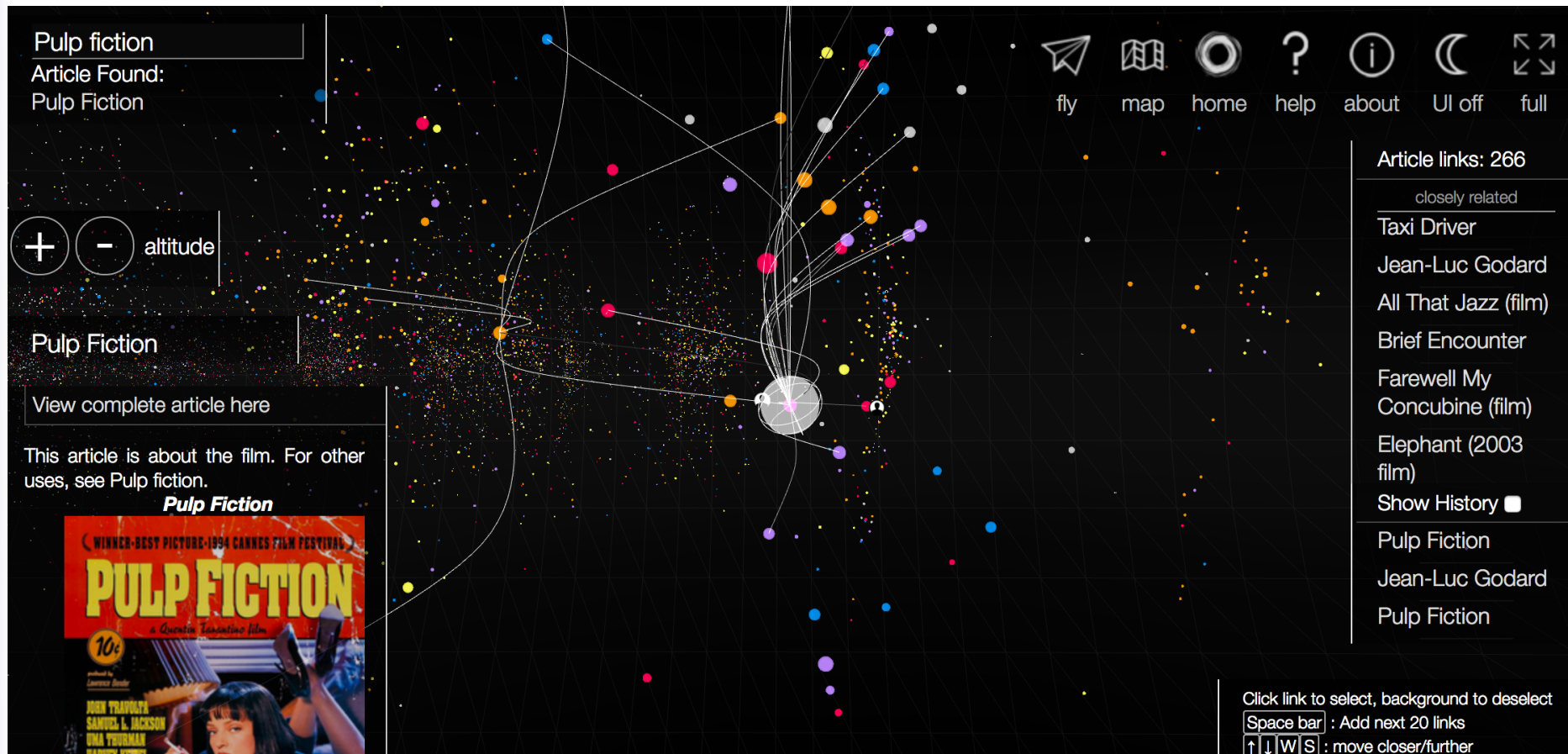
RelFinder

The screenshot displays the RelFinder web application interface. On the left, the 'between' tab is active, showing two input fields: (1) 'Kill Bill' and (2) 'Pulp Fiction (film)'. Below these are 'add', 'clear', and 'Find Relations' buttons. A 'Filter by:' section shows 'length' selected, with a table displaying 'number of objects' (2), 'num' (8/8), and 'vi'. The 'examples' tab is also visible. The main area on the right shows a semantic network graph. Nodes include 'Uma Thurman', 'Miramax Films', 'Kill Bill', 'Pulp Fiction', 'Lawrence Bender', and 'Quentin Tarantino'. Edges represent relations like 'starring', 'distributor', 'producer', 'writer', 'director', and 'wikiPageRe...'. The 'Kill Bill' node is highlighted with a black border, and 'Pulp Fiction (film)' is highlighted with a blue border. The status bar at the top right indicates 'Status: Idle'.

Philipp Heim, Steffen Lohmann and Timo Stegemann. Interactive Relationship Discovery via the Semantic Web. In: Proceedings of ESWC 2010, volume 6088, pages 303-317. Springer, 2010.



Exploration of Wikipedia with a Visual Interface



Exploration of Facts Relevant to a Text of Interest via Visual Interface

DaCena

Interactive exploration via visual interface

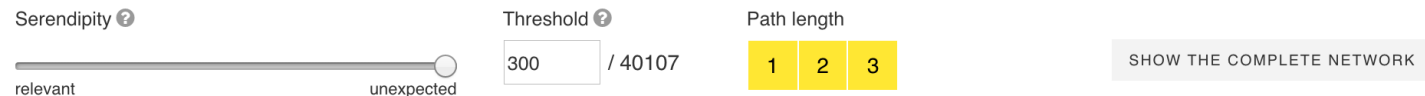
A Threat to Spanish Democracy

By CAYETANA ÁLVAREZ DE TOLEDO,
NÚRIA AMAT and MARIO VARGAS LLOSA
NOV. 7, 2014

[Original Article](#)

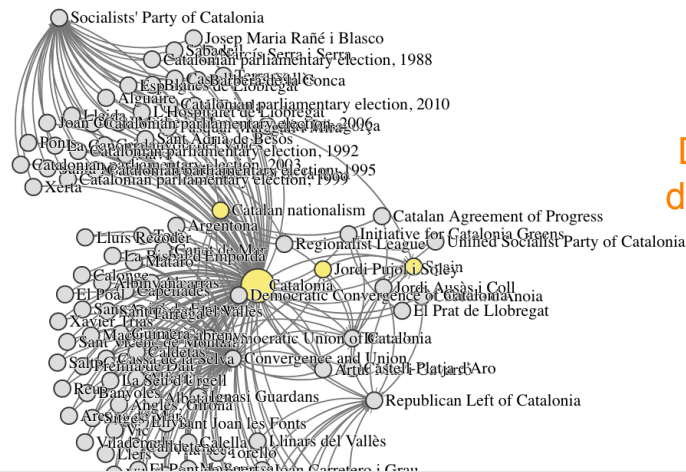
MAIN ENTITY: **CATALONIA**

On Sunday, in defiance of an order from the Spanish Constitutional Court, a farcical referendum on Catalan **separatism** will take place. It is not clear who is holding it, as there are neither official voters' rolls nor auditors, only a semiofficial cabal of volunteers. All this in the name of "democracy" and the right to decide, but without respect for the rule of law or the true will of the people, and without acknowledging the gravity of the consequences that the secession of **Catalonia** would occasion. This supposed referendum — now rebranded a



Overview

Details



Paths (i.e., semantic associations) extracted from the DBpedia KG because deemed to be *relevant* for the articles



Catalonia	leaderName	Artur Mas i Gavarró	president	Democratic Convergence of Catalonia	ideology	Catalan nationalism
Catalonia	isPartOf	Esplugues de Llobregat	leaderParty	Socialists' Party of Catalonia	ideology	Catalan nationalism
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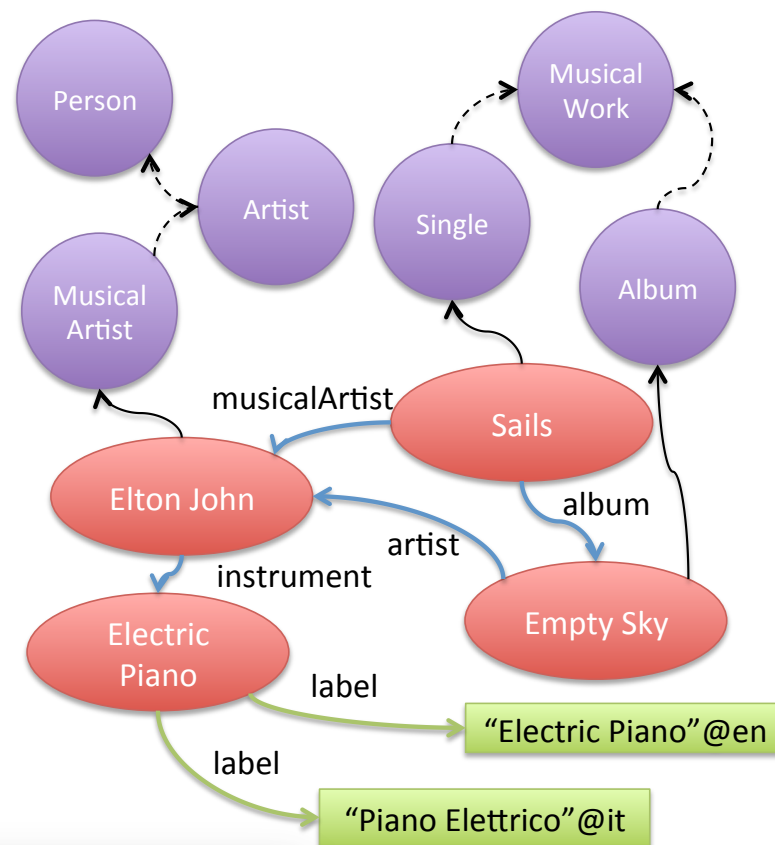


“Big” Knowledge Graphs

- DBpedia (2015): 
 - 6.2M **entities** (955K with geo coordinates)
 - 5M **entities** classified in an **ontology** (1.6M persons, 800K locations, 267K organizations)
 - 8.8B facts (triples)
 - E.g., < Barack Obama, luogo di nascita, Honolulu>

- **Ontology (schema):**

- 739 **classes**
- ~1000 **object properties**
- ~1600 **datatype properties**



```
<Electric Piano, label, "Electric Piano"@en>
<Electric Piano, label, "Piano Elettrico"@it>
<Elton John, instrument, Electric Piano>
<Empty Sky, artist, Elton John>
<Sails, musicalArtist, Elton John>
<Sails, album, Empty Sky>
<Elton John, instrument, Electric Piano>
<Elton John, type, Musical Artist>
<Sails, type, Single>
<Empty Sky, type, Album>
```

```
<Musical Artist, subClassOf, Artist>
<Artist, subClassOf, Person>
<Album, subClassOf, Musical Work>
<Single, subClassOf, Musical Work>
```



Exploration of Facts Relevant to a Text of Interest via Visual Interface

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Serendipity

Threshold

40000 / 40107

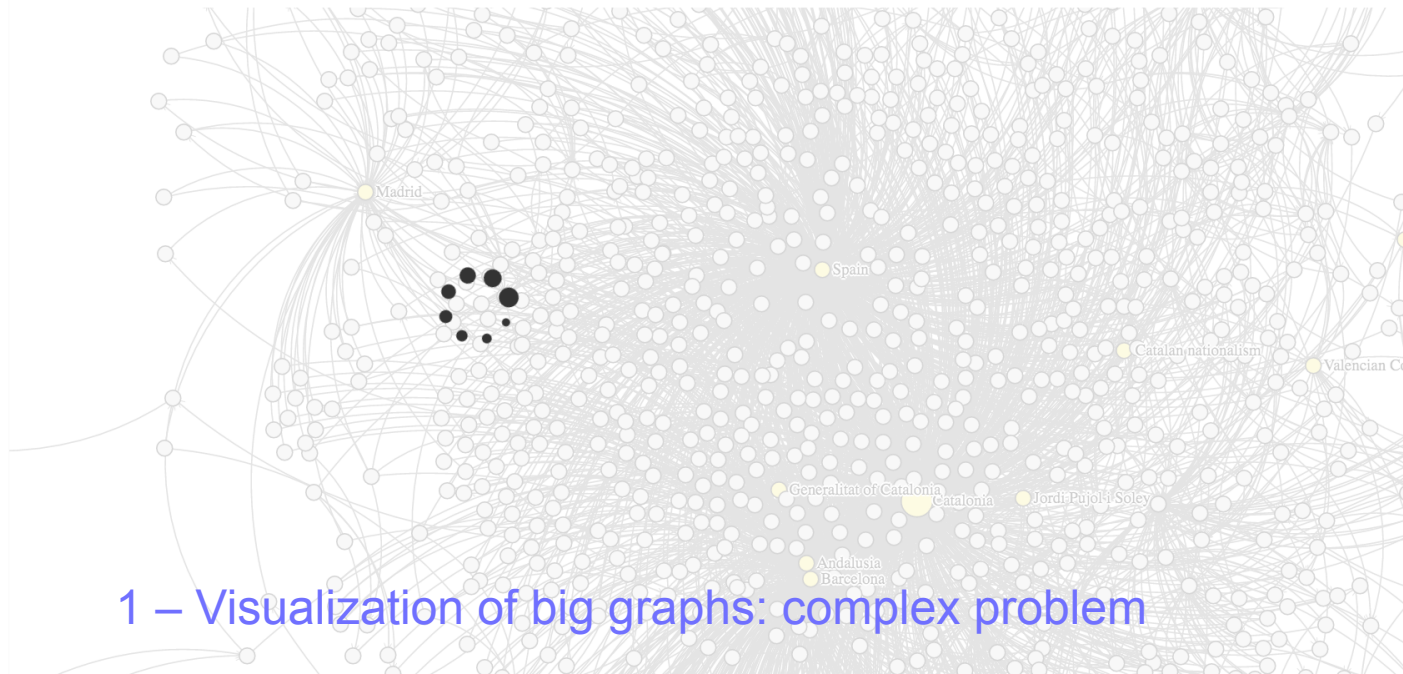
Path length

1 2 3

SHOW THE COMPLETE NETWORK

relevant

unexpected



Trying to visualize 40k associations



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Serendipity ?

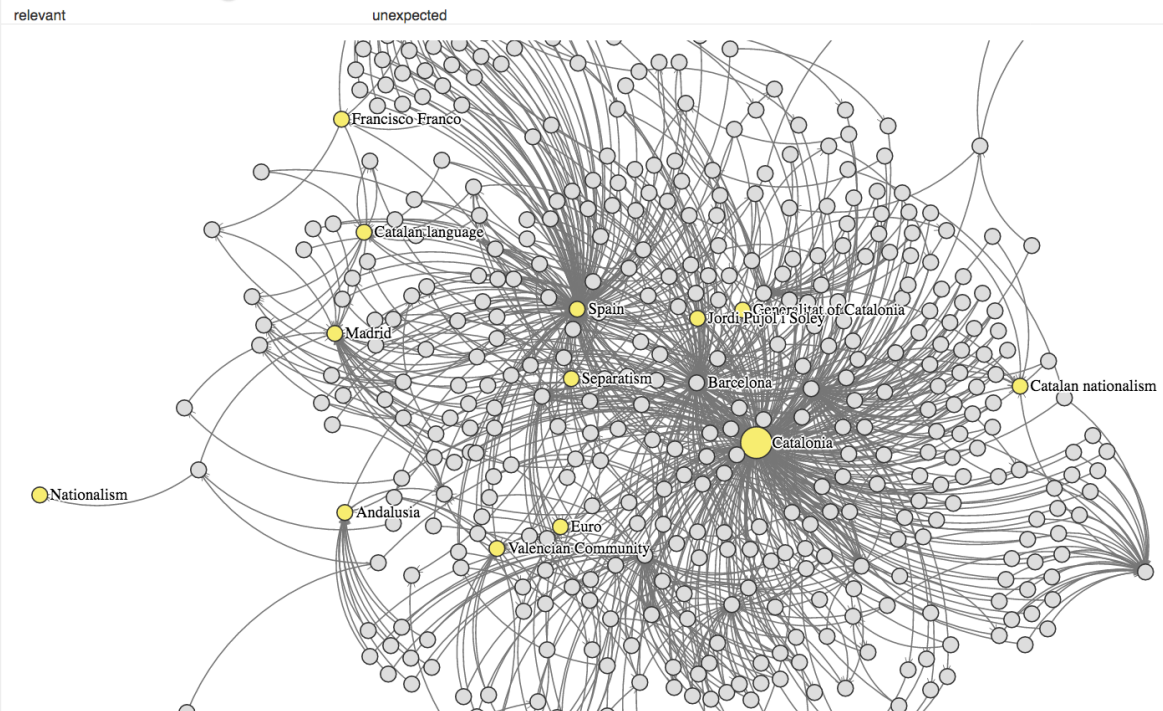
Threshold ?

Path length

1000 / 40107

1 2 3

SHOW THE COMPLETE NETWORK



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Visualizing 1K associations (of 40k)

2 – Visualization of KGs vs plain graphs
(nodes and edges have different meaning)



Abstraction and Visualization of Large Graphs

A screenshot analyzing a dataset with
about 600 millions triples
Summary view (A)



Provides **high-level contextual information**, such as:

- data size,
- current selection size, and, optionally,
- statistics (i.e., data peeking)

By C. Batini (Keynote @ICWE 2017)

P. C. Wong *et al.*, "A visual analytics paradigm enabling trillion-edge graph exploration," *2015 IEEE 5th Symposium on Large Data Analysis and Visualization (LDAV)*, Chicago, IL, 2015, pp. 57-64.



Schema-based Profiling of Large KGs to Understand their Content

ABSTAT: profiling large KGs
(e.g., DBpedia 2015-10)

How many entities of type Company?
Which types are linked to each other?

Which properties are used to describe the entities and with which frequency?

Occurrence of types (as minimal types) and properties

	subject type (occurrences)	predicate (occurrences)	object type (occurrences)	frequency	instances	Max subsj-obj	Avg subsj-obj	Min subsj-obj	Max subj-objs	Avg subj-objs	Min subj-objs
filter	dbo:Company	predicate	object								
🔍	dbo:Company (51898)	DTP foaf:name (5002938)	rdfs:Literal (12948466)	52391	64155	200	1	1	16	1	1
🔍	dbo:Company (51898)	OP foaf:homepage (362016)	owl:Thing	42861	52699	27	1	1	11	1	1
🔍	dbo:Company (51898)	OP dbo:industry (50100)	owl:Thing	42408	50093	1750	10	1	15	1	1
🔍	dbo:Company (51898)	DTP dbo:foundingYear (87412)	xmls:gYear (3469462)	42131	49085	1159	99	1	8	1	1
🔍	dbo:Company (51898)	OP dbo:predecessor (78232)	dbo:Person (611330)	8	11	1	1	1	1	1	1

Minimal Patterns: there exist entities that have Company as *minimal type*, which are linked to literals that have gYear as *minimal type* by the property foundingYear

Cardinality descriptors: max/avg/min number of different subjects associated with a same object (and vice versa)

Frequency and instances: how many times this pattern occurs as minimal type pattern and as a non minimal type pattern (considers frequency of more specific patterns)



Which Entities and Relations are More Interesting?

More popular...

Less popular ... (more unexpected?)

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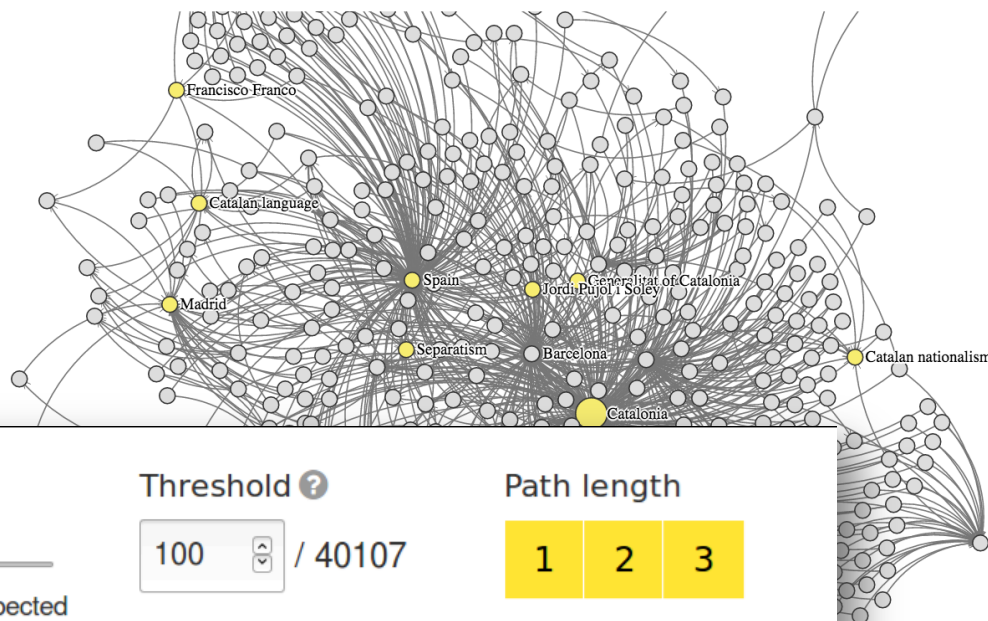
Threshold ?

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Path length

1 2 3

SHOW THE COMPLETE NETWORK



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Serendipity ?

Threshold ?

Path length

relevant

unexpected

100 / 40107

1 2 3

Separatism

Heuristics: **serendipity**, i.e., entities rilevanti wrt the article but links via unfrequent properties



But different users are interested in different information! (Bianchi&al, 2017)

KG Profiling + Machine Learning

- Let's profile entities and relations with a set of features

- Topological features (saliency, etc.):

- PageRank (Page&al, 1999)
 - HITS (Kleinberg&al, 1999)

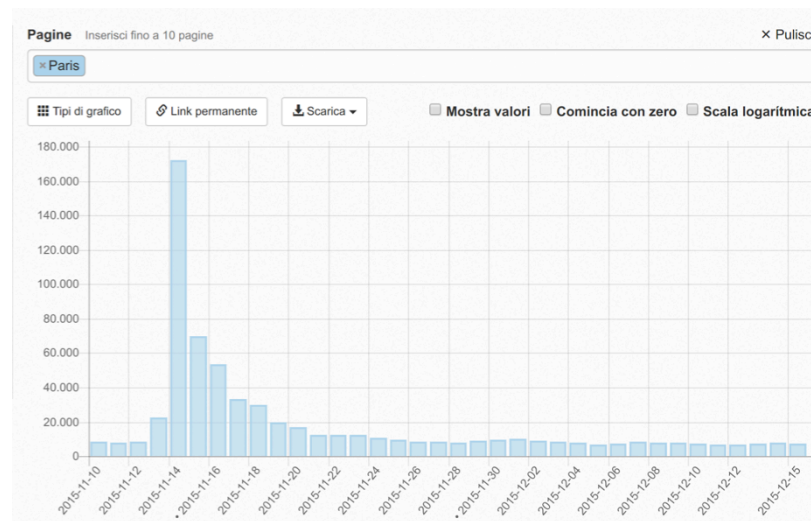
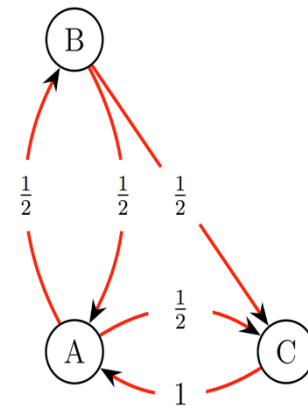
- Features of relevance:

- Relevance (Palmonari&al, 2015),
 - Temporal Relevance (Bianchi&al, 2017)

- ...

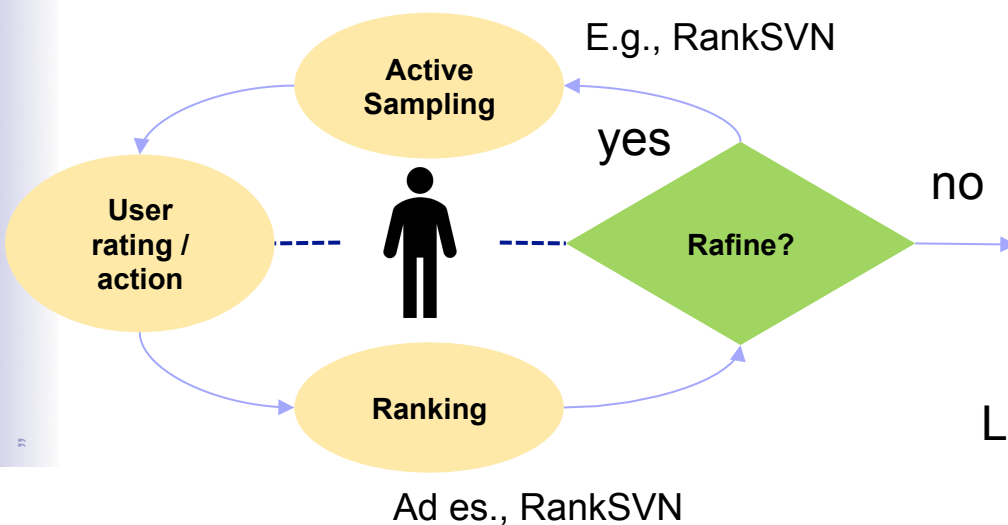
- ...

... Or features based on entity similarity, e.g., NLP (e.g., Cogito) + deep learning (Bianchi&al, 2018; DiCarlo&al, 2019)



KG Profiling + Machine Learning

- Let's profile entities and relations with a set of features
 - ... (see previous slide)
 -
- Let's learn which features make the graph elements (semantic associations in this case) more relevant for the individual user
 - **Explicit rating**, clicks, navigation patterns



Ranking by “estimated interest”

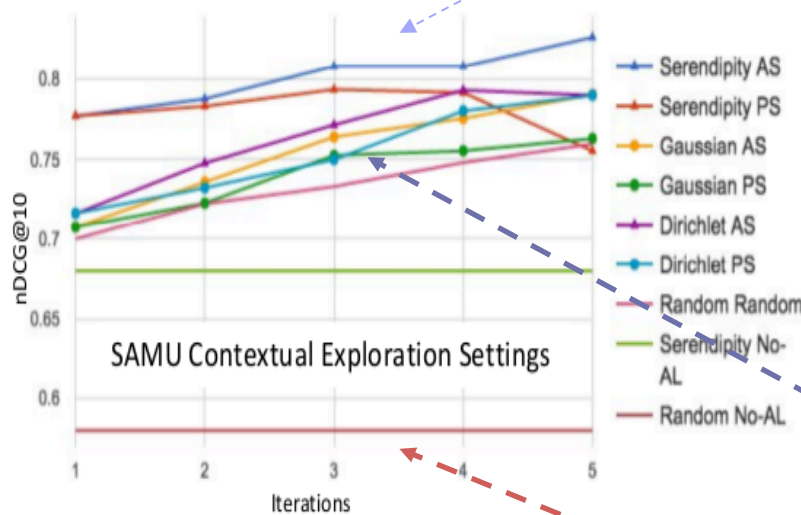
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Learning loop for ranking information

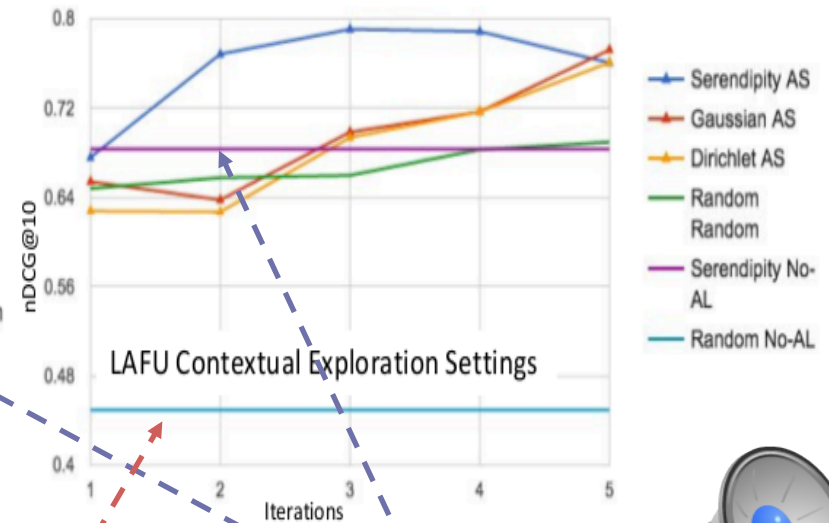
Experimental Results: Learning with Explicit Rating

Heuristics (serendipity) + active learning

→ Quality of the ranking (nDCG)



→ Iterations
(# user ratings)

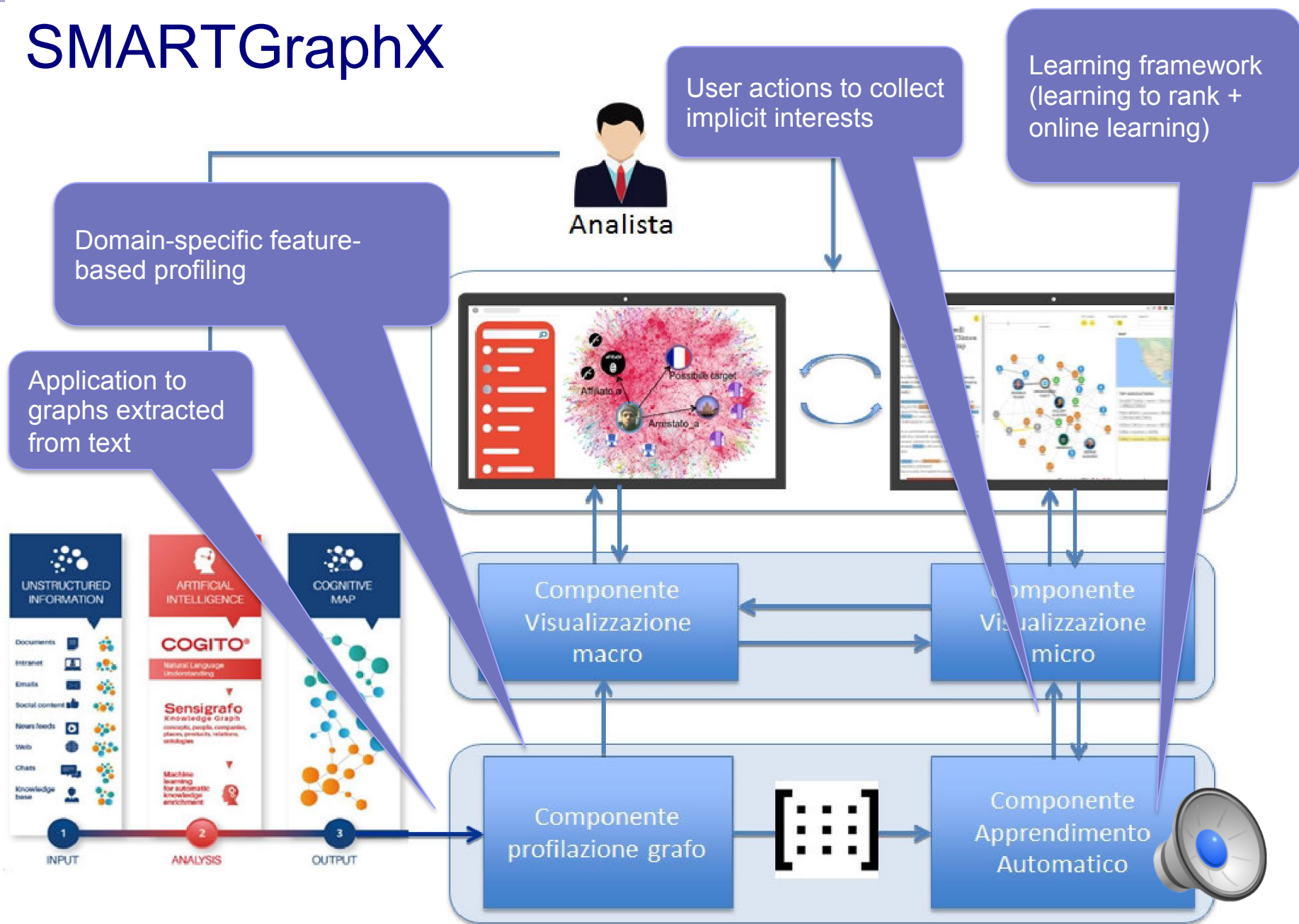


Random ranking

Heuristics (serendipity)
without learning



SMARTGraphX



References



Ontology-based KG profiling...

- Renzo Arturo Alva Principe, Blerina Spahiu, Matteo Palmonari, Anisa Rula, Flavio De Paoli, Andrea Maurino: **ABSTAT 1.0: Compute, Manage and Share Semantic Profiles of RDF Knowledge Graphs**. ESWC (Satellite Events) 2018: 170-175
- Blerina Spahiu, Riccardo Porrini, Matteo Palmonari, Anisa Rula, Andrea Maurino: **ABSTAT: Ontology-Driven Linked Data Summaries with Pattern Minimalization**. ESWC (Satellite Events) 2016: 381-395
- Tommaso Di Noia, Corrado Magarelli, Andrea Maurino, Matteo Palmonari, Anisa Rula: **Using Ontology-Based Data Summarization to Develop Semantics-Aware Recommender Systems**. ESWC 2018: 128-144

Relevance and learning users' interests ...

- Federico Bianchi, Matteo Palmonari, Marco Cremaschi, Elisabetta Fersini: **Actively Learning to Rank Semantic Associations for Personalized Contextual Exploration of Knowledge Graphs**. ESWC (1) 2017: 120-135
- Matteo Palmonari, Giorgio Ubaldi, Marco Cremaschi, Daniele Ciminieri, Federico Bianchi: **DaCENA: Serendipitous News Reading with Data Contexts**. ESWC (Satellite Events) 2015: 133-137
- Valerio Di Carlo, Federico Bianchi, Matteo Palmonari: **Training Temporal Word Embeddings with a Compass**. AAAI 2019: 6326-6334
- Federico Bianchi, Matteo Palmonari, Debora Nozza: **Towards Encoding Time in Text-Based Entity Embeddings**. International Semantic Web Conference (1) 2018: 56-71
- Federico Bianchi, Matteo Palmonari: **Joint Learning of Entity and Type Embeddings for Analogical Reasoning with Entities**. NL4AI@AI*IA 2017: 57-68